z SMARTBRIDGE | SKILL WALLET

Coding & Solution

Date	06-07-2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository / GitHub Link	https://github.com/gmanideep81-source/OptiCrop
Programming Language(s)	Python
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Crop prediction using ML (Logistic Regression), K-Means clustering for exploratory analysis, soil & climate input form, multi-page web interface (Home, About, Find Your Crop, Result)
Pending / Incomplete Features	None
Setup / Run Instructions	pip install -r requirements.txt python app.py Open http://localhost:5000 in your browser Live deployment: https://opticrop-407a.onrender.com

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes